Endpoint–Bickley Reduction for Frequency-Independent Low-Froude Evaluation of Michell Wave Resistance

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Abstract

Michell's thin-ship formula remains attractive for rapid, geometry-sensitive wave-resistance prediction, but its outer integral becomes increasingly oscillatory as the length Froude number decreases. Conventional real-axis marching must resolve a growing number of bow–stern oscillations and its tail termination can become unreliable. For hulls represented by polynomial tensor-product B-splines, we show that the exact inner transform can be written as a finite sum of waves attached to spline-span endpoints. At low Froude number, terms attached below the waterline may be omitted with an explicit absolute bound. Expanding the squared transform before integration reduces the remaining resistance to pairwise kernels $K_s(\omega) = \int_1^\infty e^{i\omega\lambda}\lambda^{-s}(\lambda^2-1)^{-1/2}\,d\lambda$, which are analytic continuations of Bickley–Naylor functions. The substitution $\lambda = 1 + t^2$, followed by the exact rotation $t = e^{i\pi/4}y/\sqrt{|\omega|}$, converts each nonzero-frequency kernel to a Gaussian-decaying integral whose quadrature cost is independent of frequency. A hybrid implementation accepts the reduced result only when its omission bound and a coarse/fine contour estimate meet the requested tolerance. For the standard Wigley hull at $Fn = 0.02$, the method uses 1,152 kernel nodes instead of 511,504 inner-transform evaluations, reduces median runtime from 3.923×10^1 ms to 4.500×10^{-2} ms (approximately 880-fold), and differs from an independent real-axis reference by 1.53×10^{-13} relatively. The method is presently restricted to upright symmetric monohulls and the contour discretization estimate is not a formal interval certificate.

1 Introduction

Michell's 1898 thin-ship theory expresses wave resistance as a one-dimensional integral of the squared Fourier–Laplace transform of the longitudinal hull slope [Michell, 1898, Tuck, 1989]. It is inexpensive compared with nonlinear free-surface computation and retains the bow–stern interference that makes resistance strongly dependent on speed and hull shape. These properties have sustained its use in preliminary design, multihull studies, and mathematical shape optimization [Lazauskas, 2009, Dambrine et al., 2016].

The apparent simplicity conceals a difficult numerical limit. Let $\nu = g/U^2$, where U is speed and g gravitational acceleration. As U and therefore $Fn = U/\sqrt{gL}$ decrease, the phase $\nu\lambda x$ oscillates increasingly rapidly over the unbounded outer coordinate λ. Tuck used Filon-like longitudinal integration, while Lazauskas used exact piecewise-quadratic formulas and a fixed angular rule; the latter reports that very low Froude number remains exceptional [Tuck, 1989, Lazauskas, 2009]. A real-axis adaptive marcher must spend more work as the frequency rises and must decide when an oscillatory algebraic tail is negligible. At the pre-fix revision `64925d4`, a single quiet-window criterion reported 5.123×10^{-9} relative error at $Fn = 0.05$, while the measured error was 5.781×10^{-7}: an understatement by a factor of about 113. The frozen kernel corrects this engineering defect by advancing quiet windows on oscillatory phase alone and summing refinement and tail errors. Across $0.02 \leq Fn \leq 1$, its reported estimate covers the measured error by approximately fourfold; actual marcher error falls by factors of 3–100 and evaluation counts are roughly halved.

Low-speed endpoint dominance is not new. Keller and Ahluwalia showed that the small-Froude wave field and resistance are controlled by bow and stern waterline data [Keller and Ahluwalia, 1976]. Gotman later separated polynomial hull contributions into endpoint derivatives and bow–stern interference terms [Gotman, 2002]. Numerical steepest descent is likewise mature [Huybrechs and Vandewalle, 2006]; it has been applied to the Kelvin Green-function integral in linear ship-wave theory [Motygin, 2017]. The contribution of the present work is the following specific composition:

1. an exact, finite endpoint decomposition on every polynomial B-spline span, including interior knots and full-multiplicity chine knots;
2. an explicit bound on all resistance pairs involving an omitted submerged endpoint;
3. identification of the retained outer pair kernels as analytically continued Bickley–Naylor functions and their evaluation on a frequency-scaled Gaussian contour; and
4. conservative automatic dispatch between this reduced solver and the general real-axis method.

Each mathematical ingredient has precedent. We therefore claim a new algorithmic construction and implementation, not priority for endpoint asymptotics or steepest descent themselves. A targeted search found no prior source containing all four elements, but this negative search is not a proof of novelty; de Sendagorta and Grases [1988], whose full text was unavailable to us, is a particularly relevant uncertainty.

2 Michell resistance and spline geometry

Let $y = f(x, z) \geq 0$ denote the half-breadth of an upright symmetric hull, with $z \geq 0$ measured downward from the undisturbed waterline. In the normalization used by the implementation,

$$R_{\mathrm{w}} = \frac{4\rho g^2}{\pi U^2} \int_1^\infty \frac{\lambda^2}{\sqrt{\lambda^2 - 1}} \left|F(\lambda)\right|^2 \, d\lambda, \tag{1}$$

$$F(\lambda) = \iint_\Omega f_x(x, z) \mathrm{e}^{-\nu\lambda^2 z} \mathrm{e}^{\mathrm{i}\nu\lambda x} \, dx \, dz, \qquad \nu = \frac{g}{U^2}. \tag{2}$$

The hull is a clamped, non-rational tensor-product B-spline surface. On a knot rectangle $S = [x_0, x_1] \times [z_0, z_1]$, its longitudinal derivative is an exact bivariate polynomial,

$$f_x|_S = \sum_{a=0}^{p-1} \sum_{b=0}^{q} c_{ab}^S (x - x_0)^a (z - z_0)^b. \tag{3}$$

No station or waterline sampling is introduced: the coefficients are extracted once from exact spline derivatives at span corners. Repeated knots create zero-length intervals, which are discarded, while an interior multiplicity equal to the degree represents a continuous chine exactly.

3 Exact endpoint decomposition

The finite polynomial degree makes repeated integration by parts terminate. For a polynomial p of degree d,

$$\int_a^b p(x)\mathrm{e}^{\mathrm{i}kx}\,dx = \sum_{r=0}^{d} \frac{(-1)^r}{(\mathrm{i}k)^{r+1}} \left[p^{(r)}(x)\mathrm{e}^{\mathrm{i}kx}\right]_a^b, \tag{4}$$

$$\int_c^d p(z)\mathrm{e}^{-\kappa z}\,dz = -\sum_{u=0}^{d} \frac{1}{\kappa^{u+1}} \left[p^{(u)}(z)\mathrm{e}^{-\kappa z}\right]_c^d. \tag{5}$$

Applying equations (4) and (5) to every monomial in equation (3), with $k = \nu\lambda$ and $\kappa = \nu\lambda^2$, yields the following exact representation.

Proposition 1 (Finite endpoint representation)**.** *For a polynomial tensor-product B-spline hull with longitudinal degree at least one, the transform in equation (2) can be written*

$$F(\lambda) = \sum_{j=1}^{M} c_j \lambda^{-n_j} \mathrm{e}^{\mathrm{i}\nu\lambda x_j} \mathrm{e}^{-\nu\lambda^2 z_j}, \qquad n_j \geq 3, \tag{6}$$

where (x_j, z_j) are endpoints of nonzero knot spans and the complex coefficients c_j are finite combinations of span-polynomial endpoint derivatives. Equal endpoint/power triples may be combined exactly.

Proof. Equations equations (4) and (5) replace a monomial of derivative orders (r, u) by endpoint factors proportional to $(\nu\lambda)^{-(r+1)}(\nu\lambda^2)^{-(u+1)}$. Its power of λ is therefore $n = r + 2u + 3$. Both sums terminate at the polynomial degrees, and summing the finite contributions from all nonzero knot rectangles gives equation (6). □

The code tests equation (6) directly against its independent exact moment machinery for the Wigley hull and for a multi-span hull containing a full-multiplicity chine. Across $1 \leq \lambda \leq 20$, the discrepancy is below the test's scaled 5×10^{-10} threshold.

4 Waterline reduction with an omission bound

Partition equation (6) into waterline terms W, for which $z_j = 0$, and submerged terms S, for which $z_j > 0$. At low Froude number, every submerged term contains at least $\mathrm{e}^{-\nu z_j}$ because $\lambda \geq 1$. Retaining only W gives

$$\widetilde{F}(\lambda) = \sum_{j\in W} c_j \lambda^{-n_j} \mathrm{e}^{\mathrm{i}\nu\lambda x_j}. \tag{7}$$

Expanding $|\widetilde{F}|^2$ before outer integration is essential. It removes the nonanalytic complex conjugation from the integrand and gives

$$\widetilde{R}_{\mathrm{w}} = C \sum_{i,j\in W} \Re\Big\{c_i \overline{c_j} K_{s_{ij}}(\omega_{ij})\Big\}, \quad C = \frac{4\rho g^2}{\pi U^2}, \tag{8}$$

where $s_{ij} = n_i + n_j - 2 \geq 4$, $\omega_{ij} = \nu(x_i - x_j)$, and

$$K_s(\omega) = \int_1^\infty \frac{\mathrm{e}^{\mathrm{i}\omega\lambda}}{\lambda^s\sqrt{\lambda^2 - 1}}\,d\lambda. \tag{9}$$

Proposition 2 (Submerged-pair error bound)**.** *Let $\mathcal{P}_S = \{(i,j) : i \in S$ or $j \in S\}$. The resistance discarded by replacing F with $\widetilde{F}$ satisfies*

$$\left| R_{\mathrm{w}} - \widetilde{R}_{\mathrm{w}} \right| \le C \sum_{(i,j)\in\mathcal{P}_S} |c_i c_j| \mathrm{e}^{-\nu(z_i+z_j)} K_{s_{ij}}(0). \tag{10}$$

Proof. Expand $|F|^2 - |\widetilde{F}|^2$ into precisely the ordered pairs in $\mathcal{P}_S$. Apply the triangle inequality, use $|\mathrm{e}^{\mathrm{i}\omega\lambda}| = 1$, and bound $\mathrm{e}^{-\nu\lambda^2(z_i+z_j)} \le \mathrm{e}^{-\nu(z_i+z_j)}$ for $\lambda \ge 1$. The remaining positive integral is $K_{s_{ij}}(0)$. □

The zero-frequency kernel is analytic:

$$K_s(0) = \int_0^\infty \cosh^{-s} t \, dt = \frac{\sqrt{\pi}\,\Gamma(s/2)}{2\Gamma((s+1)/2)}, \tag{11}$$

and is evaluated by a stable even/odd recurrence.

5 Bickley kernels and Gaussian contour quadrature

With $\lambda = \cosh t$, equation (9) becomes

$$K_s(\omega) = \int_0^\infty \mathrm{e}^{\mathrm{i}\omega\cosh t} \cosh^{-s} t \, dt = \mathrm{Ki}_s(-\mathrm{i}\omega), \tag{12}$$

where the final equality is understood by analytic continuation of the Bickley–Naylor function [Bickley and Nayler, 1935, NIST Digital Library of Mathematical Functions, 2026]. This identification supplies recurrences and possible special-function implementations, although the present solver evaluates the contour directly.

The square-root endpoint in equation (9) is first removed by $\lambda = 1 + t^2$:

$$K_s(\omega) = 2\mathrm{e}^{\mathrm{i}\omega} \int_0^\infty \frac{\mathrm{e}^{\mathrm{i}\omega t^2}}{(1+t^2)^s\sqrt{2+t^2}} \, dt. \tag{13}$$

For $\omega > 0$, analyticity in the intervening sector permits the exact rotation $t = \mathrm{e}^{\mathrm{i}\pi/4} y/\sqrt{\omega}$, giving

$$K_s(\omega) = \frac{2\mathrm{e}^{\mathrm{i}(\omega+\pi/4)}}{\sqrt{\omega}} \int_0^\infty \frac{\mathrm{e}^{-y^2} \, dy}{(1+\mathrm{i}y^2/\omega)^s\sqrt{2+\mathrm{i}y^2/\omega}}. \tag{14}$$

Negative frequencies follow by conjugation. The oscillation is gone, and the frequency dependence now varies slowly in the amplitude. This is the same general numerical-asymptotic principle that yields frequency-independent work in high-frequency scattering [Gibbs et al., 2020], specialized here to a simple phase with a closed contour.

The implementation applies 24- and 48-point Gauss–Legendre rules on $0 \le y \le 8$. The neglected Gaussian tail obeys

$$|K_s - K_s^{(8)}| \le \frac{\mathrm{e}^{-64}}{8\sqrt{2\omega}}, \qquad \omega > 0, \tag{15}$$

using $|1 + \mathrm{i}y^2/\omega| \ge 1$, $|\sqrt{2 + \mathrm{i}y^2/\omega}| \ge \sqrt{2}$, and the standard Gaussian-tail bound. At the minimum supported nonzero frequency $\omega = 25$, this is below 4×10^{-30}. The 24/48 difference estimates discretization error. That difference is empirical, not a proof; the method therefore has a rigorous submerged-term bound and an effectively negligible analytic contour tail, but not a fully certified total floating-point error.

For fixed geometry, every nonzero endpoint pair costs 72 scalar nodes, independent of $\nu|x_i - x_j|$; self pairs use equation (11) without quadrature. The Wigley test case requires 1,152 scalar nodes at every supported low Froude number.

6 Hybrid algorithm and implementation scope

The reduced method is an asymptotic companion, not a blind replacement for the general solver. The public algorithm is:

1. validate the physical conditions and exact spline geometry;
2. form and combine the exact endpoint terms in equation (6);
3. reject the specialization unless the hull is upright, symmetric, has its spline domain beginning exactly at $z = 0$, and has longitudinal degree at least one;
4. reject it if any distinct retained endpoint pair has $0 < |\omega_{ij}| < 25$;
5. evaluate equation (8), the bound equation (10), and the weighted 24/48 contour difference; and
6. accept the result only if their combined relative estimate is no greater than the caller's requested tolerance. Otherwise run the general real-axis marcher.

The frequency gate avoids claiming asymptotic efficiency before oscillation is strong enough for the fixed contour policy. More importantly, the error gate uses the actual hull, speed, and resistance cancellation: a low nominal Froude number alone is not sufficient. At the default relative tolerance 10^{-5}, the Wigley reduction is rejected at $Fn = 0.08$ and accepted at $Fn = 0.05$.

The implementation is in safe Rust and uses no external numerical dependency. The reduced solver is tested independently at three levels: exact inner decomposition, isolated Bickley kernels, and total resistance. The general validation harness additionally covers Froude similarity, zero camber for symmetric sides, classical catamaran $4\cos^2$ interference, tolerance self-convergence, and reported versus observed error.

7 Numerical experiments

7.1 Independent reference

The standard Wigley half-breadth is

$$f(x,z) = \frac{B}{2}\left(1 - \frac{4x^2}{L^2}\right)\left(1 - \frac{z^2}{T^2}\right), \quad |x| \le L/2, \quad 0 \le z \le T, \tag{16}$$

with $L = 1.000 \times 10^1$ m, $B = 1.000$ m, and $T = 6.250 \times 10^{-1}$ m. The independent reference does not call the production moment or outer-integral code. It evaluates the analytic Wigley inner transform and applies 16-point Gauss–Legendre panels on the positive real λ axis with Kahan-compensated accumulation. Panel boundaries advance the bow phase by at most $\pi/2$, and the integration is continued to $\lambda = 4000$. At $Fn = 0.02$, doubling the cutoff to 8000 changes the reference by 1.23×10^{-15} relatively; floating-point effects remain, so this cutoff check is a resolution diagnostic rather than an interval bound.

Table 1 shows two regimes. At $Fn = 0.08$, omitted submerged terms are still material and the estimate correctly rejects the reduction at default tolerance. At $Fn \le 0.05$, the reduced/reference differences are between 1.53×10^{-13} and 1.03×10^{-12}, within the reduced solver's empirical estimates. The general marcher's error grows as Froude number falls. At $Fn = 0.05$, its corrected reported

Table 1: Resistance validation against the independent analytic-Wigley, resolved-real-axis reference. "Reduced estimate" combines the rigorous submerged-pair bound and empirical contour difference.

Fn	Reference R_{w} (N)	Reduced rel. diff.	Reduced estimate	Marcher rel. diff.
0.08	1.946×10^{-1}	1.172×10^{-5}	3.814×10^{-5}	6.918×10^{-8}
0.05	1.058×10^{-2}	2.070×10^{-13}	3.728×10^{-12}	1.493×10^{-7}
0.03	5.114×10^{-4}	1.034×10^{-12}	3.462×10^{-12}	3.322×10^{-7}
0.02	4.417×10^{-5}	1.526×10^{-13}	7.620×10^{-13}	6.334×10^{-7}

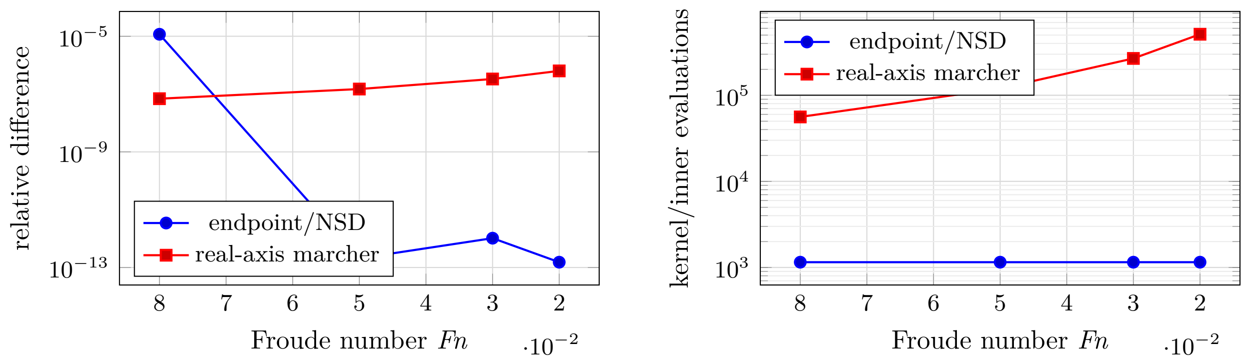

Figure 1: Observed accuracy (left) and work (right) as Froude number decreases. The deepened independent reference resolves endpoint/NSD differences down to the 10^{-13}–10^{-12} level, while its quadrature work is fixed by geometry.

relative error is 5.774×10^{-7}, compared with the measured 1.493×10^{-7}, so the estimate covers rather than understates the observed error. This corrected, independently checked marcher is the comparison method throughout table 1.

The isolated kernel tests compare equation (14) against dense real-axis integration for $s \in \{4, 7, 10\}$ and $|\omega| \in \{25, 100, 400\}$, with scaled discrepancy below 10^{-9}. The complete Rust workspace contains 215 passing test cases, including one documentation test; the companion Python interface has 11 passing tests.

7.2 Runtime

Timing used an optimized build, 30 warmed samples, and the default relative tolerance 10^{-5}. Absolute timings are host-dependent; the work counts and checksums are deterministic. The two $Fn = 0.02$ methods were measured in the same benchmark process.

At $Fn = 0.02$, the median speedup is approximately 880-fold. The endpoint solver's cost is nearly unchanged between $Fn = 0.05$ and 0.02, as predicted by equation (14). The ordinary 21-speed design sweep remains on whichever path satisfies the error gate. Its checksum is $2.553251156101 \times 10^{5}$, an intentional change from the pre-fix value because the corrected marcher retains a previously truncated positive tail.

8 Relation to prior work and scope of novelty

The endpoint structure has a long lineage. The small-Froude analysis of Keller and Ahluwalia [1976] expresses leading resistance and waves through bow and stern waterline slopes. Gotman [2002] repeatedly differentiates polynomial hull functions, separates endpoint self terms from bow–stern

Table 2: Final release benchmark on the development host.

Case	Median (ms)	Best (ms)	Diagnostic
21 speeds, $Fn = 0.10 \ldots 0.50$	26.289	22.843	corrected checksum
Default API, $Fn = 0.05$	0.050	0.050	1,152 nodes
Real-axis marcher, $Fn = 0.02$	39.233	35.950	511,504 evaluations
Endpoint/NSD, $Fn = 0.02$	0.045	0.044	1,152 nodes

interference, and discusses higher endpoint derivatives. The present finite B-spline decomposition should be viewed as a general, implementation-ready form of that known mechanism, not a new physical asymptotic principle.

Similarly, exact or Filon-like integration of piecewise-polynomial inner data is established practice [Tuck, 1989, Lazauskas, 2009]. The special feature here is that exact endpoint reduction continues through the nonlinear-looking outer modulus square: analytic pair expansion converts the outer problem into a small set of reusable special-function kernels. The relation $K_s = \mathrm{Ki}_s(-\mathrm{i}\omega)$ appears not to have been exploited in the inspected Michell literature. Very recent work gives finite Bessel–Struve module representations for integer-order Bickley functions [Ruffa and Toni, 2026]; its stability on the imaginary axis may support an even faster future kernel implementation.

Contour deformation is well established for oscillatory quadrature [Huybrechs and Vandewalle, 2006]. Motygin [2017] applies steepest descent and Clenshaw–Curtis quadrature in ship-wave theory, but to the oscillatory part of the Kelvin Green function rather than Michell's resistance integral. Automated methods now handle general phases and coalescing saddles [Gibbs et al., 2024]. Such machinery is a promising route to multihull transverse phases, but is unnecessary for the present quadratic phase.

Our literature search included publisher and DOI records, arXiv, TRID, the Adelaide repository, DLMF, and backward references in the sources above. It did not find a previous method combining arbitrary-degree B-spline endpoint terms, Bickley analytic continuation, frequency-scaled Gaussian quadrature, and a submerged-endpoint error gate for Michell resistance. However, the search was not exhaustive, no patent search was performed, and only the abstract of de Sendagorta and Grases [1988] was available. The defensible claim is therefore a possibly new algorithmic combination, subject to expert literature review.

9 Limitations and open extensions

The current evidence establishes a sharp numerical advantage within a narrow model, not a universal low-speed ship-wave solution.

- **Geometry.** The reduced solver supports upright symmetric single hulls. Asymmetric, heeled, and multihull paths fall back to the general method.

- **Error certification.** The submerged-pair omission bound and Gaussian tail bound are analytic. The 24/48 quadrature difference and floating-point roundoff are not interval-certified.

- **Reference floor.** The independent real-axis reference uses compensated summation through $\lambda = 4000$; at $Fn = 0.02$, a cutoff-doubling check changes it by 1.23×10^{-15}. Differences near the observed 10^{-13} scale should still not be read as interval-certified digits.

- **Physical model.** Michell theory is linear, inviscid, slender, deep-water, and fixed-attitude. At very low speed, viscous effects may dominate the already small wave resistance. Nonlinear low-Froude waves can be beyond all algebraic orders and controlled by Stokes phenomena; that different problem is not resolved by better quadrature [Trinh and Chapman, 2013].

The highest-leverage numerical extension is multihull support. A transverse placement introduces phase $\nu y\lambda\sqrt{\lambda^2-1}$, with moving saddles absent from equation (14); a uniform or automated contour treatment is required. Retaining shallow submerged endpoints similarly produces mixed linear and quadratic complex phases. Other priorities are interval-certified contour errors, stable imaginary-argument Bickley recurrences, finite-depth dispersion, and differentiation of the endpoint representation for low-Froude hull optimization.

10 Conclusions

Polynomial B-spline geometry permits more than exact evaluation of Michell's inner transform. By carrying integration by parts to its finite endpoint form, expanding the squared transform into analytic endpoint pairs, and recognizing their outer integrals as continued Bickley functions, the low-Froude problem can be moved from an increasingly oscillatory real axis to a fixed Gaussian contour. The resulting work is determined by geometry rather than Froude number. An explicit submerged-pair bound makes the approximation testable and allows a conservative hybrid solver to fall back outside its useful regime.

On the standard Wigley case, this construction is both more accurate and almost three orders of magnitude faster than the existing low-Froude march. The result is a strong engineering advance and a possibly new combination of known ideas. Broader geometry, fully certified error control, and independent review of the novelty claim remain necessary before presenting it as a general state-of-the-art replacement.

Reproducibility statement

The implementation, exact endpoint and kernel tests, independent Wigley reference, benchmark harness, and research report are contained in the local `michell` repository at revision `64925d4` and subsequent paper commits. Before public submission, this statement should be replaced by a permanent archive DOI.

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